

8085 Addressing Modes & Interrupts

Now let us discuss the addressing modes in 8085 Microprocessor.

Addressing Modes in 8085

These are the instructions used to transfer the data from one register to another register, from the memory to the register, and from the register to the memory without any alteration in the content. Addressing modes in 8085 is classified into 5 groups –

Immediate addressing mode

In this mode, the 8/16-bit data is specified in the instruction itself as one of its operand.

For example: MVI K, 20F: means 20F is copied into register K.

Register addressing mode

In this mode, the data is copied from one register to another. **For example:** MOV K, B: means data in register B is copied to register K.

Direct addressing mode

In this mode, the data is directly copied from the given address to the register. **For example:** LDB 5000K: means the data at address 5000K is copied to register B.

Indirect addressing mode

In this mode, the data is transferred from one register to another by using the address pointed by the register. **For example:** MOV K, B: means data is transferred from the memory address pointed by the register to the register K.

Implied addressing mode

This mode doesn't require any operand; the data is specified by the opcode itself. **For example:** CMP.